Dynamic Modelling for the Efficient Power and Heat Management of Polymer Electrolyte Fuel Cell System

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Abstract

The real time control of fuel cell vehicle using a well established technology of model predictive control requires accurate mathematical models. The dynamic phenomenon taking place inside fuel cell involves complex interaction of mass transport, energy transport and electrochemical kinetics. The complexity further increases when the fuel cell dynamics is coupled with the vehicle transmission dynamics. These uncertain dynamics determine the overall operational performance of the system. In addition, the real time control requires computing machinery to run the optimisation algorithms. This further increases manufacturing and operational costs. In order to resolve these issues the main goals of this paper are (a) develop the dynamic model of integrated system and (b) design explicit model predictive controller for the efficient operation of overall system. Such a modelling and control framework is critical to the effective operation and preventive maintenance of the system. We conclude that the proposed "model predictive control on a chip" strategy is the best available option for making smarter decisions in operating fuel cells vehicles reliability and further improving their competitive advantage.

Keywords: Fuel cell vehicle, DC motor, multi-parametric optimisation, model predictive control.

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1 Introduction

Fuel cell is the most promising energy enabling technology for future with hydrogen envisioned as the sustainable energy carrier. The polymer electrolyte membrane fuel cell is the most widely researched fuel cell type. Due to its low operating temperatures and higher efficiency compared to the internal combustion engine it has found widespread adoption in many areas including the transport sector. Thanks to the global policy drivers and the extensive research carried out over last few decades, the entry of commercial fuel cell vehicles has witnessed an unprecedented growth with optimistic market forecast. In order to reach the targeted market penetration, fuel cell system integration and smart control automation are the key research objectives that can improve safety, reliability, performance, and overall economy of the fuel cell vehicles. Although the fuel cell stack technology has reached a level of maturity, systemwide control automation for better electricity and thermal management still remains the main research priorities.

The real time control of fuel cell vehicle using a well established technology of model predictive control requires accurate mathematical models. The dynamic phenomenon taking place inside fuel cell involves complex interaction of mass transport, energy transport and electrochemical kinetics. The complexity further increases when the fuel cell dynamics is coupled with the

*The financial support from EPSRC grants (EP/I014640/1 and EP/K503381/1) and sponsorships from ARM Incorporation is gratefully acknowledged. Corresponding author: amit@imperial.ac.uk

vehicle transmission dynamics. These uncertain dynamics determine the overall operational performance of the system. In addition, the real time control requires computing machinery to run the optimisation algorithms. This further increases manufacturing and operational costs. In order to resolve these issues the main goals of this paper are (a) develop the dynamic model of integrated system and (b) design explicit model predictive controller for the efficient operation of overall system.

To achieve these goals, we first presents a combined dynamic model for the vehicle transmission system and the open cathode polymer electrolyte membrane fuel cell. Next, using the developed model we carry out worst case uncertainty analysis to determine the optimal operating regime. Finally, by employing multi-parametric programming techniques we design an explicit model predictive controller for the efficient thermal management of the fuel cell powertrain. The resulting map of control policies can be easily embedded on a low cost microcontroller and deployed onboard fuel cell vehicle. We describe step by step deployment procedure of our embedded controller onboard a mini vehicle vehicle. Our in silico analysis and hardware-in-the-loop experimental driving cycle results demonstrate that significant reduction in operational costs can be achieved while simultaneously improving overall performance the vehicle. Such a modelling and control framework is critical to the effective operation and preventive maintenance of the system. Thus, we conclude that our proposed "model predictive control on a chip" strategy is the best available option for making smarter decisions in operating fuel cells vehicles reliability and further improving their competitive advantage.

2 Theoretical Developments

This section presents the theoretical developments. First we develop a dynamic model of fuel cell system and later vehicle powertrain. In following section we build the dynamic model of the fuel cell.

2.1 Dynamics of fuel cell

Tyre Deformation when Rolling

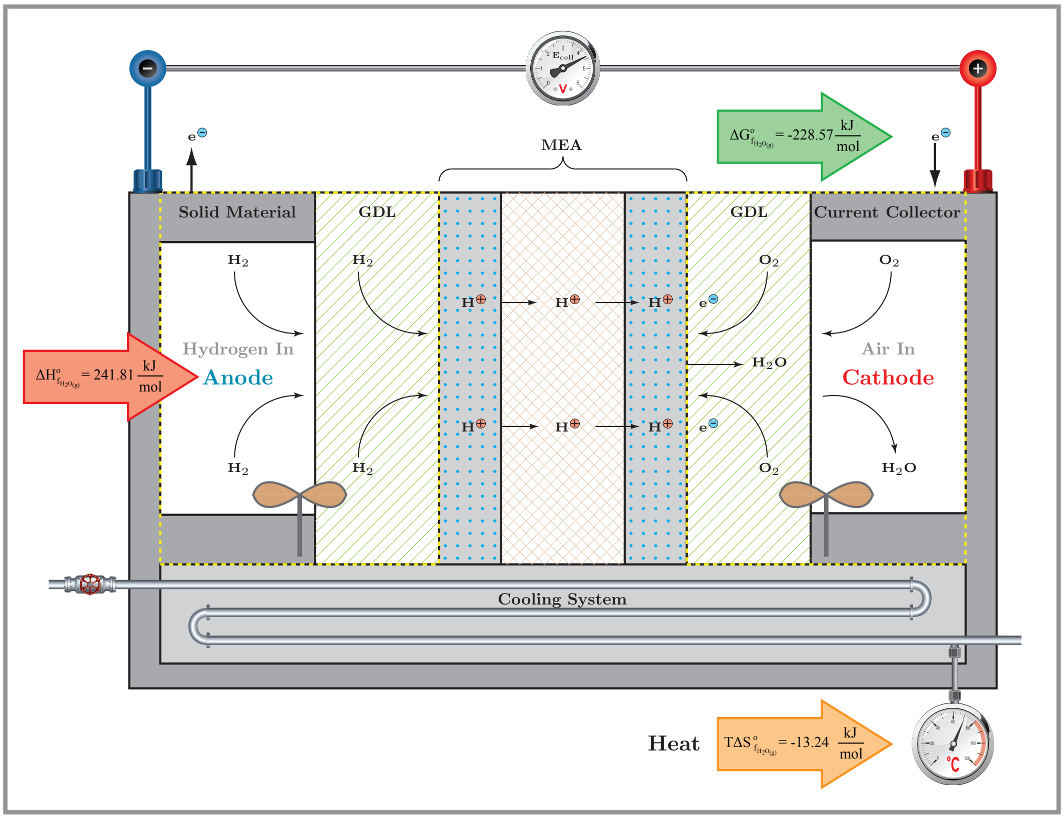

Figure 1: Simplified schematic illustrating the inner working principle of unit cell

In this work we have adapted a model for fuel cell stack which is made up of number of unit cells connected in series. The unit cells are made up of a polymer electrolyte membrane sandwiched between anode and cathode electrodes while each of the gas diffusion layer and the flow chamber is represented by a continuous stirred tank reactor (CSTR), [1]. A simplified schematic illustrating inner workings of this unit cell is shown in Figure 4. The electrochemical reactions involved in the process are described in following equations.

$$\begin{aligned} \text{Anode}: \quad & \mathrm{H_2} && \longrightarrow \mathrm{2H^+} \;+\; \mathrm{2e^-} && (1)\\ \text{Cathode}: \quad & \mathrm{2H^+} + \tfrac{1}{2}\mathrm{O_2} + \mathrm{2e^-} && \longrightarrow \mathrm{H_2O} && (2)\\ \text{Overall}: \quad & \mathrm{H_2} + \tfrac{1}{2}\mathrm{O_2} && \longrightarrow \mathrm{H_2O} \;+\; \text{Heat} \;+\; \text{Electricity} && (3) \end{aligned}$$

2.1.1 Assumptions

We assume that the gases are ideal. The stack is fed with pure hydrogen at anode and it reacts quickly and completely. At cathode oxygen is taken from the air supplied by the fan with atmospheric air molar composition O_2:N_2 = 21:79. The cathode exhaust contains water as vapor, excess oxygen and nitrogen from as inert gas. At cathode temperature is stable and constant at all time due to uniform mixing.

2.1.2 Mole balances

In this section we develop the dynamic model of fuel cell with balance of plant. The rate of generation of water per unit area of the membrane is given by

$$r_{H_2O} = \frac{j}{nF} \tag{4}$$

where j is the current density, F is the Faraday's constant, and n is the number of electrons taking part in the electrochemical reaction. Due to assumptions made we neglect the material balance on the anode side and only focus our attention on the cathode chamber with overall reaction. With the notations described in nomenclature, the species mole balances results into following equations.

$$V_{cat}\frac{dC_{O_2}}{dt} = C_{O_2}^{in}F_{cat}^{in} - C_{O_2}F_{cat}^{out} - \frac{1}{2}r_{H_2O}A_{mem} \tag{5}$$

$$V_{cat}\frac{dC_{H_2O}}{dt} = C_{H_2O}^{in}F_{cat}^{in} - C_{H_2O}F_{cat}^{out} + r_{H_2O}A_{mem} \tag{6}$$

$$V_{cat}\frac{dC_{N_2}}{dt} = C_{N_2}^{in}F_{cat}^{in} - C_{N_2}F_{cat}^{out} \tag{7}$$

2.1.3 Mass balance at cathode

Assuming there is no accumulation of overall mass the expression for cathodic outflow is given by

$$\left(C_{O_2} + C_{N_2} + C_{H_2O}\right)F_{cat}^{out} = \left(C_{O_2}^{in} + C_{N_2}^{in} + C_{H_2O}^{in}\right)F_{cat}^{in} + \frac{1}{2}r_{H_2O}A_{mem} \tag{8}$$

2.1.4 Energy balance at cathode

Due to uniform mixing temperature dynamics at cathode is neglected resulting into following expression

$$T = \frac{F_{cat}^{in}T_{amb} + UA_{cat}T_{sol}}{F_{cat}^{out} + UA_{cat}} \tag{9}$$

2.1.5 Overall Energy balance

The energy generated in the form of heat is given by

$$Q_h = -\Delta H_{f_{H_2O_g}} r_{H_2O} - P_e \tag{10}$$

where temperature dependent heat of formation of water is

$$\Delta H_{f_{H_2O_g}} = \Delta H^o_{f_{H_2O_g}} + \int_{T_o}^{T} C_p dT \tag{11}$$

Substituting for heat capacities of species with empirical relation $C_p = c_1 + c_2T + c_3T^2 + c_4T^3$ we get,

$$\Delta H_{f_{H_2O_g}} = \Delta H^o_{f_{H_2O_g}} + \sum_{i=1}^{4}\frac{1}{i}\Delta c_i\left(T^i - T_o^i\right) \tag{12}$$

where $\Delta c_i = c_i^{H_2O} - c_i^{H_2} - \frac{1}{2}c_i^{O_2}$.

2.1.6 Fuel cell electrochemistry

The free energy of formation of water is given by

$$\Delta G_{f_{H_2O}} = \Delta G^o_{f_{H_2O_g}} + RT\ln\left(\frac{a_{H_2O}}{a_{H_2}\sqrt{a_{O_2}}}\right) \tag{13}$$

where ΔG_f and ΔG_f^o are the free energies available for electrical work at operating and standard conditions ; a_{H_2}, a_{O_2} and a_{H_2O} are the activities of the species involved which are for ideal gas equal to the concentration of the species; R is the gas constant; T is the operating temperature. Note that we are using the lower value since water formed is assumed to be in vapor form. With free energy change $\Delta G = -nFE$ where E is the reversible potential and F is Faraday's constant, equation (13) leads to following Nernst equation with E^o as the open circuit potential,

$$E_{ner} = E^o - \frac{RT}{nF} \ln \left(\frac{C_{H_2O}}{C_{H_2} \sqrt{C_{O_2}}} \right) \tag{14}$$

During the operation cell undergoes various overpotential losses which are given by,

$$E_{cell} = E_{ner} - E_{act} - E_{ohm} - E_{con} \tag{15}$$

At low current densities the activation loss is experienced due to the slow kinetics of electrochemical reaction creating barrier to the flow of electrons and reversal transfer of charge. The activation polarization is represented by

$$E_{act} = \frac{RT}{\alpha nF} \ln \left(\frac{j + j_{loss}}{j_o} \right) \tag{16}$$

where loss in current j_{loss} is assumed to be zero and j_o is the exchange current density. The temperature dependent exchange current density j_o is given by

$$j_o = nFk_r e^{(-\beta nFE^o/RT)} \tag{17}$$

where k_r is the associated with the speed of reaction. The next loss is potential is due to the internal electronic resistance through electrodes and external circuit as well as the ionic resistance in electrolyte. It varies proportionally to the increase in current given by Ohm's law $E_{ohm} = jR_{ohm}$. The internal resistance of the cell is due to interplay of temperature and current density by following empirical relationship

$$R_{ohm} = r_1 + r_2 T + r_3 j. \tag{18}$$

Although occurs at all operating conditions but prominent at higher current densities the drop in potential is due to mass transfer effect also known as concentration loss where reactant species finds difficulty in reaching active catalyst sites. This polarization effect is given by following empirical relation.

$$E_{con} = \frac{\gamma RT}{nF} e^{nj} \tag{19}$$

where γ is a coefficient determining limiting current density.

Finally, the electrical power generated is simply $P_e = jE_{cell}$.

2.2 Energy Balance at System Boundary B1

$$\frac{dT_{V81}}{dt} = \frac{r_{F81} F_{H_2}}{V_{V81}} T_{H_2}^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}} V_{V81}} T_{V31} - \frac{r_{F81} F_{H_2}}{V_{V81}} T_{V81} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}} V_{V81}} \tag{20}$$

where the split ratio r_{F81} at Tee TF-81 is

$$r_{F81} \leq \begin{cases} 1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 1 \\ 0.5 + My_{F81} & \text{if} \quad y_{F81} = 0 \end{cases} \tag{21}$$

$$-r_{F81} \leq \begin{cases} -1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 1 \\ -0.5 + My_{F81} & \text{if} \quad y_{F81} = 0 \end{cases} \tag{22}$$

2.3 Energy Balance at System Boundary B2

$$\frac{dT_{E82}}{dt} = \frac{r_{F81} F_{H_2}}{V_{E82}} T_{V81} + \frac{(1 - r_{F81}) F_{H_2}}{V_{E82}} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}} V_{E82}} (T_{E82} - T_a) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}} V_{E82}} \tag{23}$$

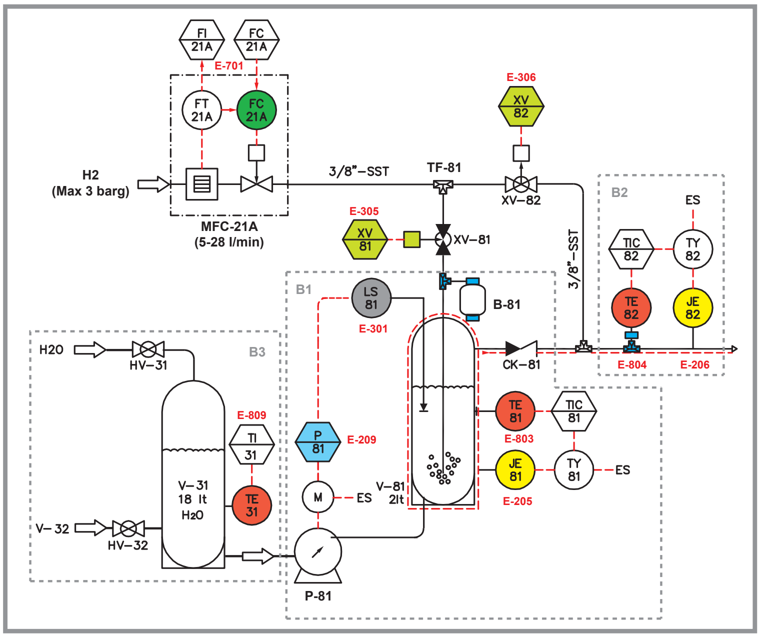

Figure 2: Humidification system for hydrogen

2.4 Energy Balance at System Boundary B3

$$\frac{dT_{V31}}{dt} = \frac{F_W^{in}}{V_{V31}} T_W^{in} + \frac{F_{W_{V32}}}{V_{V31}} T_{V32} - \frac{F_{W_{V31}}}{V_{V31}} T_{V31} \tag{24}$$

2.5 Energy Balance at System Boundary B4

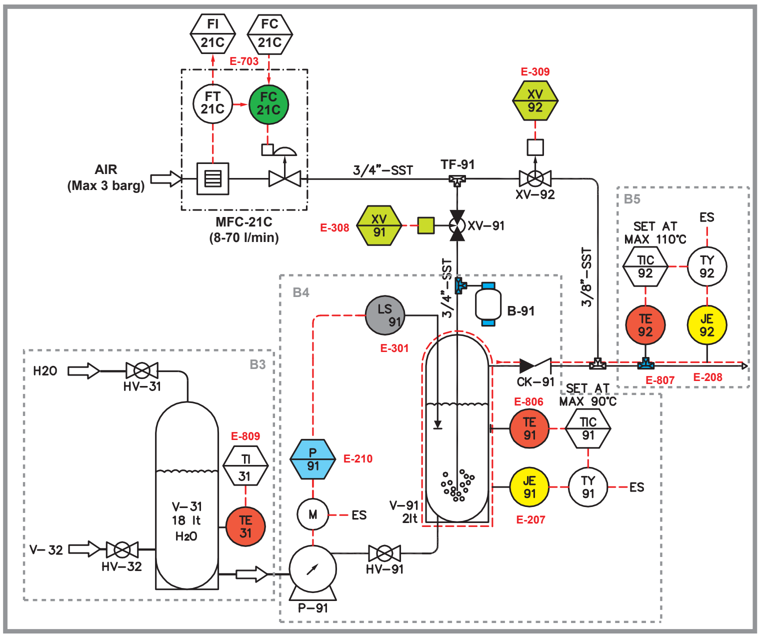

Figure 3: Humidification system for air

$$\frac{dT_{V91}}{dt} = \frac{r_{F91} F_A}{V_{V91}} T_A^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_A C_{p_A} V_{V91}} T_{V31} - \frac{r_{F91} F_A}{V_{V91}} T_{V91} + \frac{Q_{V91}}{\rho_A C_{p_A} V_{V91}} \tag{25}$$

where the split ratio r_{F91} at Tee TF-91 is

$$r_{F91} \leq \begin{cases} 1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 1 \\ 0.5 + M y_{F91} & \text{if} \quad y_{F91} = 0 \end{cases} \tag{26}$$

$$-r_{F91} \leq \begin{cases} -1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 1 \\ -0.5 + M y_{F91} & \text{if} \quad y_{F91} = 0 \end{cases} \tag{27}$$

2.6 Energy Balance at System Boundary B5

$$\frac{dT_{E92}}{dt} = \frac{r_{F91} F_A}{V_{E92}} T_{V91} + \frac{(1 - r_{F91}) F_A}{V_{E92}} T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A} V_{E92}} (T_{E92} - T_a) + \frac{Q_{JE92}}{\rho_A C_{p_A} V_{E92}} \tag{28}$$

2.7 Energy Balance at System Boundary B6

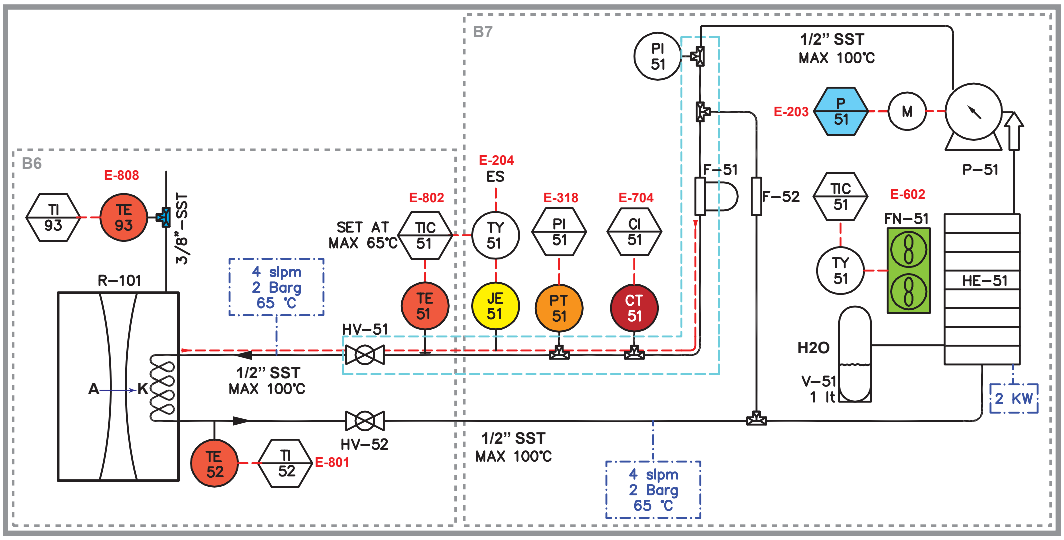

Figure 4: Cooling system with radiator fan and water circulation

$$\frac{dT_{E52}}{dt} = \frac{F_{CW}}{V_{CW}} T_{E51} - \frac{F_{CW}}{V_{CW}} T_{E52} + \frac{U_f A_f}{\rho_W C_{p_W} V_{CW}} (T_{E93} - T_{E51}) \tag{29}$$

2.8 Energy Balance at System Boundary B7

$$\frac{dT_{E51}}{dt} = \frac{F_{CW}}{V_R} T_{E52} - \frac{F_{CW}}{V_R} T_{E51} - \frac{U_r A_r}{\rho_W C_{p_W} V_R} (T_{E52} - T_a) S_{FN51} \tag{30}$$

3 Energy Balance

$$\frac{dT_{V81}}{dt} = \frac{r_{F81}F_{H_2}}{V_{V81}}T_{H_2}^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}} V_{V81}}T_{V31} - \frac{r_{F81}F_{H_2}}{V_{V81}}T_{V81} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}} V_{V81}} \quad (31)$$

$$\frac{dT_{E82}}{dt} = \frac{r_{F81}F_{H_2}}{V_{E82}}T_{V81} + \frac{(1-r_{F81})\,F_{H_2}}{V_{E82}}T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}} V_{E82}}\left(T_{E82}-T_a\right) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}} V_{E82}} \quad (32)$$

$$\frac{dT_{V31}}{dt} = \frac{F_W^{in}}{V_{V31}}T_W^{in} + \frac{F_{W_{V32}}}{V_{V31}}T_{V32} - \frac{F_{W_{V31}}}{V_{V31}}T_{V31} \quad (33)$$

$$\frac{dT_{V91}}{dt} = \frac{r_{F91}F_A}{V_{V91}}T_A^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_A C_{p_A} V_{V91}}T_{V31} - \frac{r_{F91}F_A}{V_{V91}}T_{V91} + \frac{Q_{V91}}{\rho_A C_{p_A} V_{V91}} \quad (34)$$

$$\frac{dT_{E92}}{dt} = \frac{r_{F91}F_A}{V_{E92}}T_{V91} + \frac{(1-r_{F91})\,F_A}{V_{E92}}T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A} V_{E92}}\left(T_{E92}-T_a\right) + \frac{Q_{JE92}}{\rho_A C_{p_A} V_{E92}} \quad (35)$$

$$\frac{dT_{E52}}{dt} = \frac{F_{CW}}{V_{CW}}T_{E51} - \frac{F_{CW}}{V_{CW}}T_{E52} + \frac{U_f A_f}{\rho_W C_{pw} V_{CW}}\left(T_{E93}-T_{E51}\right) \quad (36)$$

$$\frac{dT_{E51}}{dt} = \frac{F_{CW}}{V_R}T_{E52} - \frac{F_{CW}}{V_R}T_{E51} - \frac{U_r A_r}{\rho_W C_{pw} V_R}\left(T_{E52}-T_a\right)S_{FN51} \quad (37)$$

$$r_{F81} \leq \begin{cases} 1.0 + M(1-y_{F81}) & \text{if} \quad y_{F81}=1 \\ 0.5 + My_{F81} & \text{if} \quad y_{F81}=0 \end{cases} \quad (38)$$

$$-r_{F81} \leq \begin{cases} -1.0 + M(1-y_{F81}) & \text{if} \quad y_{F81}=1 \\ -0.5 + My_{F81} & \text{if} \quad y_{F81}=0 \end{cases} \quad (39)$$

$$r_{F91} \leq \begin{cases} 1.0 + M(1-y_{F91}) & \text{if} \quad y_{F91}=1 \\ 0.5 + My_{F91} & \text{if} \quad y_{F91}=0 \end{cases} \quad (40)$$

$$-r_{F91} \leq \begin{cases} -1.0 + M(1-y_{F91}) & \text{if} \quad y_{F91}=1 \\ -0.5 + My_{F91} & \text{if} \quad y_{F91}=0 \end{cases} \quad (41)$$

3.1 Fuel cell electrochemistry

$$E_{cell} = E_{ner} - E_{act} - E_{ohm} - E_{con} \quad (42)$$

$$E_{ner} = E^o - \frac{R\,T}{n\,F}\ln\left(\frac{C_{H_2O}}{C_{H_2}\sqrt{C_{O_2}}}\right) \quad (43)$$

$$E_{act} = \frac{RT}{\alpha nF}\ln\left(\frac{j + j_{loss}}{j_o}\right) \quad (44)$$

$$j_o = \kappa\, nFe^{(-\beta\ nFE^o/RT)} \quad (45)$$

$$E_{ohm} = jR_{ohm} \quad (46)$$

$$R_{ohm} = \gamma_1 + \gamma_2 T + \gamma_3 j \quad (47)$$

$$E_{con} = \frac{\zeta RT}{nF}e^{\theta j} \quad (48)$$

4 DAE Model of the Plant

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \quad \xrightleftharpoons[r_{H_2O}]{-\Delta H_{f_{H_2O_g}}} \quad \mathrm{H_2O_g} \ + \ \mathrm{Q_h} \ + \ \mathrm{P_e} \tag{49}$$

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_c^{in} - C_{O_2} F_c^{out} - \frac{1}{2} r_{H_2O} A_m \tag{50}$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} F_c^{in} - C_{H_2O} F_c^{out} + r_{H_2O} A_m \tag{51}$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_c^{in} - C_{N_2} F_c^{out} \tag{52}$$

$$\frac{dT_{E93}}{dt} = F_A^{in} T_A^{in} - F_c^{out} T_{E93} + \frac{Q_h A_m}{\rho_A C_{p_A} V_c} - \frac{U_c A_c}{\rho_A C_{p_A} V_c} (T - T_s) \tag{53}$$

$$\frac{dT_{V81}}{dt} = \frac{r_{F81} F_{H_2}}{V_{V81}} T_{H_2}^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}} V_{V81}} T_{V31} - \frac{r_{F81} F_{H_2}}{V_{V81}} T_{V81} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}} V_{V81}} \tag{54}$$

$$\frac{dT_{E82}}{dt} = \frac{r_{F81} F_{H_2}}{V_{E82}} T_{V81} + \frac{(1 - r_{F81}) F_{H_2}}{V_{E82}} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}} V_{E82}} (T_{E82} - T_a) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}} V_{E82}} \tag{55}$$

$$\frac{dT_{V31}}{dt} = \frac{F_W^{in}}{V_{V31}} T_W^{in} + \frac{F_{W_{V32}}}{V_{V31}} T_{V32} - \frac{F_{W_{V31}}}{V_{V31}} T_{V31} \tag{56}$$

$$\frac{dT_{V91}}{dt} = \frac{r_{F91} F_A}{V_{V91}} T_A^{in} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_A C_{p_A} V_{V91}} T_{V31} - \frac{r_{F91} F_A}{V_{V91}} T_{V91} + \frac{Q_{V91}}{\rho_A C_{p_A} V_{V91}} \tag{57}$$

$$\frac{dT_{E92}}{dt} = \frac{r_{F91} F_A}{V_{E92}} T_{V91} + \frac{(1 - r_{F91}) F_A}{V_{E92}} T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A} V_{E92}} (T_{E92} - T_a) + \frac{Q_{JE92}}{\rho_A C_{p_A} V_{E92}} \tag{58}$$

$$\frac{dT_{E52}}{dt} = \frac{F_{CW}}{V_{CW}} T_{E51} - \frac{F_{CW}}{V_{CW}} T_{E52} + \frac{U_f A_f}{\rho_W C_{pw} V_{CW}} (T_{E93} - T_{E51}) \tag{59}$$

$$\frac{dT_{E51}}{dt} = \frac{F_{CW}}{V_R} T_{E52} - \frac{F_{CW}}{V_R} T_{E51} - \frac{U_r A_r}{\rho_W C_{pw} V_R} (T_{E52} - T_a) S_{FN51} \tag{60}$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} P_e \tag{61}$$

$$r_{H_2O} = \frac{j}{nF} N_c \tag{62}$$

$$F_c^{out} = \frac{\left(C_{O_2}^{in} + C_{N_2}^{in} + C_{H_2O}^{in}\right) F_c^{in} + \frac{1}{2} r_{H_2O} A_m}{C_{O_2} + C_{N_2} + C_{H_2O}} \tag{63}$$

$$C_{p_A} = \frac{C_{N_2} C_p^{N_2} + C_{O_2} C_p^{O_2} + C_{H_2O} C_p^{H_2O}}{C_{N_2} + C_{O_2} + C_{H_2O}} \tag{64}$$

$$C_p^{\psi} = c_1^{\psi} + c_2^{\psi} T + c_3^{\psi} T^2 + c_4^{\psi} T^3; \qquad \forall \psi = \{H_2,\ O_2,\ N_2,\ H_2O\} \tag{65}$$

$$\Delta H_{f_{H_2O_g}} = \Delta H^o_{f_{H_2O_g}} + \sum_{i=1}^{4} \left\{ \frac{1}{i} \left(c_i^{H_2O} - c_i^{H_2} - \frac{1}{2} c_i^{O_2} \right) \left(T^i - T_o^i \right) \right\} \tag{66}$$

$$Q_e = \gamma_1 j + \gamma_2 j T + \gamma_3 j^2 \tag{67}$$

$$Q_h = -\Delta H_{f_{H_2O_g}} r_{H_2O} - P_e + Q_e \tag{68}$$

$$E_c = \left(\frac{\alpha - \beta}{\alpha} \right) E^o - \frac{RT}{nF} \left[\ln \left(\frac{1}{\sqrt{C_{O_2}}} \right) + \frac{1}{\alpha} \ln \left(\frac{j}{\kappa n F} \right) + \zeta e^{\theta j} \right] - Q_e \tag{69}$$

$$r_{F81} \leq \begin{cases} 1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 0 \\ 0.5 + M y_{F81} & \text{if} \quad y_{F81} = 1 \end{cases} \tag{70}$$

$$-r_{F81} \leq \begin{cases} -1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 0 \\ -0.5 + M y_{F81} & \text{if} \quad y_{F81} = 1 \end{cases} \tag{71}$$

$$r_{F91} \leq \begin{cases} 1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 0 \\ 0.5 + M y_{F91} & \text{if} \quad y_{F91} = 1 \end{cases} \tag{72}$$

$$-r_{F91} \leq \begin{cases} -1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 0 \\ -0.5 + M y_{F91} & \text{if} \quad y_{F91} = 1 \end{cases} \tag{73}$$

5 Dynamic Model of the Plant

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \quad \underset{r_{H_2O}}{\overset{-\Delta H_{f_{H_2O_g}}}{\rightleftharpoons}} \quad \mathrm{H_2O_g} \ + \ \mathrm{Q_h} \ + \ \mathrm{P_e} \tag{74}$$

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_A - C_{O_2} F_c^{out} - \frac{1}{2}\frac{j}{nF} N_c A_m \tag{75}$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} F_A - C_{H_2O} F_c^{out} + \frac{j}{nF} N_c A_m \tag{76}$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_c^{in} - C_{N_2} F_c^{out} \tag{77}$$

$$V_c \frac{dT_{E93}}{dt} = F_A T_A^{in} - F_c^{out} T_{E93} + \frac{Q_h A_m}{\rho_A C_{p_A}} - \frac{U_c A_c}{\rho_A C_{p_A}} \left(T_{E93} - T_{E52}\right) \tag{78}$$

$$V_{V81} \frac{dT_{V81}}{dt} = r_{F81} F_{H_2} T_{H_2}^{in} - r_{F81} F_{H_2} T_{V81} + \frac{\rho_W C_{pW} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}}} T_{V31} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}}} \tag{79}$$

$$V_{E82} \frac{dT_{E82}}{dt} = r_{F81} F_{H_2} T_{V81} + \left(1 - r_{F81}\right) F_{H_2} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}}} \left(T_{E82} - T_a\right) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}}} \tag{80}$$

$$V_{V31} \frac{dT_{V31}}{dt} = F_W^{in} T_W^{in} + F_{W_{V32}} T_{V32} - F_{W_{V31}} T_{V31} \tag{81}$$

$$V_{V91} \frac{dT_{V91}}{dt} = r_{F91} F_A T_A^{in} - r_{F91} F_A T_{V91} + \frac{\rho_W C_{pW} F_{W_{V31}}}{\rho_A C_{p_A}} T_{V31} + \frac{Q_{V91}}{\rho_A C_{p_A}} \tag{82}$$

$$V_{E92} \frac{dT_{E92}}{dt} = r_{F91} F_A T_{V91} + \left(1 - r_{F91}\right) F_A T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A}} \left(T_{E92} - T_a\right) + \frac{Q_{JE92}}{\rho_A C_{p_A}} \tag{83}$$

$$V_{CW} \frac{dT_{E52}}{dt} = F_{CW} T_{E51} - F_{CW} T_{E52} + \frac{U_f A_f}{\rho_W C_{pW}} \left(T_{E93} - T_{E51}\right) \tag{84}$$

$$V_R \frac{dT_{E51}}{dt} = F_{CW} T_{E52} - F_{CW} T_{E51} - \frac{U_r A_r}{\rho_W C_{pW}} \left(T_{E52} - T_a\right) S_{FN51} \tag{85}$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} P_e \tag{86}$$

$$F_c^{out} = F_c^{in} + \frac{1}{2}\frac{j}{nF} N_c A_m \tag{87}$$

$$C_{p_A} = \frac{C_{N_2} C_p^{N_2} + C_{O_2} C_p^{O_2} + C_{H_2O} C_p^{H_2O}}{C_{N_2} + C_{O_2} + C_{H_2O}} \tag{88}$$

$$C_p^{\psi} = c_1^{\psi} + c_2^{\psi} T + c_3^{\psi} T^2 + c_4^{\psi} T^3; \qquad \forall \psi = \{H_2,\ O_2,\ N_2,\ H_2O\} \tag{89}$$

$$\Delta H_{f_{H_2O_g}} = \Delta H_{f_{H_2O_g}}^{o} + \sum_{i=1}^{4} \left\{ \frac{1}{i} \left(c_i^{H_2O} - c_i^{H_2} - \frac{1}{2} c_i^{O_2} \right) \left(T^i - T_o^i \right) \right\} \tag{90}$$

$$Q_e = \gamma_1 j + \gamma_2 j T + \gamma_3 j^2 \tag{91}$$

$$Q_h = \left(-\Delta H_{f_{H_2O_g}} \right) \frac{j}{nF} N_c - P_e + Q_e \tag{92}$$

$$E_c = E^o - \frac{RT}{n\,F} \ln \left(\frac{C_{H_2O}}{C_{H_2}\,\sqrt{C_{O_2}}} \right) - \frac{RT}{\alpha n F} \ln \left(\frac{j + j_{loss}}{j_o} \right) - \frac{\zeta RT}{nF} e^{\theta j} - Q_e \tag{93}$$

$$j_o = \kappa\, nFe^{(-\beta\, nFE^o/RT)} \tag{94}$$

(95)

$$r_{F81} \leq \begin{cases} 1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 0 \\ 0.5 + My_{F81} & \text{if} \quad y_{F81} = 1 \end{cases} \tag{96}$$

$$-r_{F81} \leq \begin{cases} -1.0 + M(1 - y_{F81}) & \text{if} \quad y_{F81} = 0 \\ -0.5 + My_{F81} & \text{if} \quad y_{F81} = 1 \end{cases} \tag{97}$$

$$r_{F91} \leq \begin{cases} 1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 0 \\ 0.5 + My_{F91} & \text{if} \quad y_{F91} = 1 \end{cases} \tag{98}$$

$$-r_{F91} \leq \begin{cases} -1.0 + M(1 - y_{F91}) & \text{if} \quad y_{F91} = 0 \\ -0.5 + My_{F91} & \text{if} \quad y_{F91} = 1 \end{cases} \tag{99}$$

6 Dynamic Model of the Plant Constant Specific Heat

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \quad \underset{r_{H_2O}}{\overset{-\Delta H_{f_{H_2O_g}}}{\rightleftharpoons}} \quad \mathrm{H_2O_g} \ + \ \mathrm{Q_h} \ + \ \mathrm{P_e} \tag{100}$$

$$\begin{aligned}
V_c \frac{dC_{O_2}}{dt} &= C_{O_2}^{in} F_A - C_{O_2} F_c^{out} - \frac{1}{2}\frac{N_c A_m}{nF} j && (101)\\
V_c \frac{dC_{H_2O}}{dt} &= C_{H_2O}^{in} F_A - C_{H_2O} F_c^{out} + \frac{N_c A_m}{nF} j && (102)\\
V_c \frac{dC_{N_2}}{dt} &= C_{N_2}^{in} F_A - C_{N_2} F_c^{out} && (103)\\
V_c \frac{dT_{E93}}{dt} &= F_A T_A^{in} - F_c^{out} T_{E93} + \frac{Q_h A_m}{\rho_A C_{p_A}} - \frac{U_c A_c}{\rho_A C_{p_A}} \left(T_{E93} - T_{E52}\right) && (104)\\
V_{V81} \frac{dT_{V81}}{dt} &= r_{F81} F_{H_2} T_{H_2}^{in} - r_{F81} F_{H_2} T_{V81} + \frac{\rho_W C_{p_W} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}}} T_{V31} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}}} && (105)\\
V_{E82} \frac{dT_{E82}}{dt} &= r_{F81} F_{H_2} T_{V81} + \left(1 - r_{F81}\right) F_{H_2} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}}} \left(T_{E82} - T_a\right) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}}} && (106)\\
V_{V31} \frac{dT_{V31}}{dt} &= F_W^{in} T_W^{in} + F_{W_{V32}} T_{V32} - F_{W_{V31}} T_{V31} && (107)\\
V_{V91} \frac{dT_{V91}}{dt} &= r_{F91} F_A T_A^{in} - r_{F91} F_A T_{V91} + \frac{\rho_W C_{p_W} F_{W_{V31}}}{\rho_A C_{p_A}} T_{V31} + \frac{Q_{V91}}{\rho_A C_{p_A}} && (108)\\
V_{E92} \frac{dT_{E92}}{dt} &= r_{F91} F_A T_{V91} + \left(1 - r_{F91}\right) F_A T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A}} \left(T_{E92} - T_a\right) + \frac{Q_{JE92}}{\rho_A C_{p_A}} && (109)\\
V_{CW} \frac{dT_{E52}}{dt} &= F_{CW} T_{E51} - F_{CW} T_{E52} + \frac{U_f A_f}{\rho_W C_{p_W}} \left(T_{E93} - T_{E51}\right) && (110)\\
V_R \frac{dT_{E51}}{dt} &= F_{CW} T_{E52} - F_{CW} T_{E51} - \frac{U_r A_r}{\rho_W C_{p_W}} \left(T_{E52} - T_a\right) S_{FN51} && (111)\\
\frac{dP_e}{dt} &= j E_c N_c - \frac{1}{\eta_c} P_e && (112)\\
E_c &= E^o - \frac{R\,T_{E93}}{n\,F} \ln\left(\frac{C_{H_2O}}{C_{H_2}\sqrt{C_{O_2}}}\right) - \frac{RT_{E93}}{\alpha nF} \ln\left(\frac{j + j_{loss}}{j_o}\right) - \frac{\zeta RT_{E93}}{nF} e^{\theta j} - Q_e && (113)\\
j_o &= \kappa\, nF e^{(-\beta\, nFE^o/RT_{E93})} && (114)\\
F_c^{out} &= F_c^{in} + \frac{1}{2}\frac{N_c A_m}{nF} j && (115)\\
\Delta H_{f_{H_2O_g}} &= \Delta H^o_{f_{H_2O_g}} + \left(C_p^{H_2O} - C_p^{H_2} - \frac{1}{2} C_p^{O_2}\right) \left(T_{E93} - T_o\right) && (116)\\
Q_e &= \gamma_1 j + \gamma_2 j T_{E93} + \gamma_3 j^2 && (117)\\
Q_h &= \left(-\Delta H_{f_{H_2O_g}}\right) \frac{j}{nF} N_c - P_e + Q_e && (118)\\
& && (119)
\end{aligned}$$

7 Dynamic Model of the Fuel Cell Plant

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \quad \underset{r_{H_2O}}{\overset{-\Delta H_{f_{H_2O_g}}}{\rightleftharpoons}} \quad \mathrm{H_2O_g} + \mathrm{Q_h} + \mathrm{P_e} \tag{120}$$

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_A - C_{O_2} F_c^{out} - \frac{1}{2}\frac{N_c A_m}{nF} j \tag{121}$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} F_A - C_{H_2O} F_c^{out} + \frac{N_c A_m}{nF} j \tag{122}$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_A - C_{N_2} F_c^{out} \tag{123}$$

$$V_c \frac{dT_{E93}}{dt} = F_A T_A^{in} - F_c^{out} T_{E93} + \frac{Q_h A_m}{\rho_A C_{p_A}} - \frac{U_c A_c}{\rho_A C_{p_A}} \left(T_{E93} - T_{E52}\right) \tag{124}$$

$$V_{V81} \frac{dT_{V81}}{dt} = r_{F81} F_{H_2} T_{H_2}^{in} - r_{F81} F_{H_2} T_{V81} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}}} T_{V31} + \frac{Q_{V81}}{\rho_{H_2} C_{p_{H_2}}} \tag{125}$$

$$V_{E82} \frac{dT_{E82}}{dt} = r_{F81} F_{H_2} T_{V81} + \left(1 - r_{F81}\right) F_{H_2} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}}} \left(T_{E82} - T_a\right) + \frac{Q_{JE82}}{\rho_{H_2} C_{p_{H_2}}} \tag{126}$$

$$V_{V31} \frac{dT_{V31}}{dt} = F_W^{in} T_W^{in} + F_{W_{V32}} T_{V32} - F_{W_{V31}} T_{V31} \tag{127}$$

$$V_{V91} \frac{dT_{V91}}{dt} = r_{F91} F_A T_A^{in} - r_{F91} F_A T_{V91} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_A C_{p_A}} T_{V31} + \frac{Q_{V91}}{\rho_A C_{p_A}} \tag{128}$$

$$V_{E92} \frac{dT_{E92}}{dt} = r_{F91} F_A T_{V91} + \left(1 - r_{F91}\right) F_A T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A}} \left(T_{E92} - T_a\right) + \frac{Q_{JE92}}{\rho_A C_{p_A}} \tag{129}$$

$$V_{CW} \frac{dT_{E52}}{dt} = F_{CW} T_{E51} - F_{CW} T_{E52} + \frac{U_f A_f}{\rho_W C_{pw}} \left(T_{E93} - T_{E51}\right) \tag{130}$$

$$V_R \frac{dT_{E51}}{dt} = F_{CW} T_{E52} - F_{CW} T_{E51} - \frac{U_r A_r}{\rho_W C_{pw}} \left(T_{E52} - T_a\right) S_{FN51} \tag{131}$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} P_e \tag{132}$$

$$E_c = E^o - \frac{R\,T_{E93}}{n\,F} \ln\left(\frac{C_{H_2O}}{C_{H_2}\sqrt{C_{O_2}}}\right) - \frac{RT_{E93}}{\alpha nF} \ln\left(\frac{j + j_{loss}}{j_o}\right) - \frac{\zeta RT_{E93}}{nF} e^{\theta j} - Q_e \tag{133}$$

$$j_o = \kappa\, nFe^{\left(-\beta\, nFE^o / RT_{E93}\right)} \tag{134}$$

$$F_c^{out} = F_c^{in} + \frac{1}{2}\frac{N_c A_m}{nF} j \tag{135}$$

$$\Delta H_{f_{H_2O_g}} = \Delta H_{f_{H_2O_g}}^o + \left(C_p^{H_2O} - C_p^{H_2} - \frac{1}{2} C_p^{O_2}\right)\left(T_{E93} - T_o\right) \tag{136}$$

$$Q_e = \gamma_1 j + \gamma_2 j T_{E93} + \gamma_3 j^2 \tag{137}$$

$$Q_h = \left(-\Delta H_{f_{H_2O_g}}\right) \frac{j}{nF} N_c - P_e + Q_e \tag{138}$$

8 Model Parameters of the Fuel Cell Plant

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_A - C_{O_2} F_c^{out} - \frac{1}{2}\frac{N_c A_m}{nF} \boxed{j} \tag{139}$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} \boxed{F_A} - \boxed{C_{H_2O}} F_c^{out} + \frac{N_c A_m}{nF} j \tag{140}$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_A - C_{N_2} F_c^{out} \tag{141}$$

$$V_c \frac{dT_{E93}}{dt} = F_A T_A^{in} - F_c^{out} \boxed{T_{E93}} + \frac{Q_h A_m}{\rho_A C_{p_A}} - \frac{U_c A_c}{\rho_A C_{p_A}} \left(T_{E93} - T_{E52}\right) \tag{142}$$

$$V_{V81} \frac{dT_{V81}}{dt} = r_{F81} \boxed{F_{H_2}} T_{H_2}^{in} - r_{F81} F_{H_2} T_{V81} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}}} T_{V31} + \frac{\boxed{Q_{V81}}}{\rho_{H_2} C_{p_{H_2}}} \tag{143}$$

$$V_{E82} \frac{dT_{E82}}{dt} = r_{F81} F_{H_2} T_{V81} + (1 - r_{F81}) F_{H_2} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}}} \left(T_{E82} - T_a\right) + \frac{\boxed{Q_{JE82}}}{\rho_{H_2} C_{p_{H_2}}} \tag{144}$$

$$V_{V31} \frac{dT_{V31}}{dt} = F_W^{in} T_W^{in} + F_{W_{V32}} T_{V32} - F_{W_{V31}} T_{V31} \tag{145}$$

$$V_{V91} \frac{dT_{V91}}{dt} = r_{F91} F_A T_A^{in} - r_{F91} F_A T_{V91} + \frac{\rho_W C_{pw} F_{W_{V31}}}{\rho_A C_{p_A}} T_{V31} + \frac{\boxed{Q_{V91}}}{\rho_A C_{p_A}} \tag{146}$$

$$V_{E92} \frac{dT_{E92}}{dt} = r_{F91} F_A T_{V91} + (1 - r_{F91}) F_A T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A}} \left(T_{E92} - T_a\right) + \frac{\boxed{Q_{JE92}}}{\rho_A C_{p_A}} \tag{147}$$

$$V_{CW} \frac{dT_{E52}}{dt} = \boxed{F_{CW}} T_{E51} - F_{CW} T_{E52} + \frac{U_c A_c}{\rho_W C_{pw}} \left(T_{E93} - T_{E51}\right) \tag{148}$$

$$V_R \frac{dT_{E51}}{dt} = F_{CW} T_{E52} - F_{CW} T_{E51} - \frac{U_r A_r}{\rho_W C_{pw}} \left(T_{E52} - T_a\right) \boxed{S_{FN51}} \tag{149}$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} \boxed{P_e} \tag{150}$$

$$E_c = E^o - \frac{R\,T_{E93}}{n\,F} \ln\left(\frac{C_{H_2O}}{C_{H_2}\sqrt{C_{O_2}}}\right) - \frac{RT_{E93}}{\alpha n F} \ln\left(\frac{j + j_l}{j_o}\right) - \frac{\zeta R T_{E93}}{nF} e^{\theta j} - Q_e \tag{151}$$

$$j_o = \kappa\, nFe^{\left(-\beta\, nFE^o/RT_{E93}\right)} \tag{152}$$

$$F_c^{out} = F_c^{in} + \frac{1}{2}\frac{N_c A_m}{nF} j \tag{153}$$

$$\Delta H_{f_{H_2O_g}} = \Delta H^o_{f_{H_2O_g}} + \left(C_p^{H_2O} - C_p^{H_2} - \frac{1}{2} C_p^{O_2}\right)\left(T_{E93} - T_o\right) \tag{154}$$

$$Q_e = \gamma_1 j + \gamma_2 j T_{E93} + \gamma_3 j^2 \tag{155}$$

$$Q_h = \left(-\Delta H_{f_{H_2O_g}}\right) \frac{j}{nF} N_c - P_e + Q_e \tag{156}$$

9 Coordinated Model Parameters of the Fuel Cell Plant

$$V_c \frac{dC_{O_2}}{dt} = C_{O_2}^{in} F_A - C_{O_2} F_c^{out} - \frac{1}{2}\frac{N_c A_m}{nF} \boxed{j} \tag{157}$$

$$V_c \frac{dC_{H_2O}}{dt} = C_{H_2O}^{in} \boxed{F_A} - \boxed{C_{H_2O}} F_c^{out} + \frac{N_c A_m}{nF} j \tag{158}$$

$$V_c \frac{dC_{N_2}}{dt} = C_{N_2}^{in} F_A - C_{N_2} F_c^{out} \tag{159}$$

$$V_c \frac{dT_{E93}}{dt} = F_A T_A^{in} - F_c^{out} \boxed{T_{E93}} + \frac{Q_h A_m}{\rho_A C_{p_A}} - \frac{U_c A_c}{\rho_A C_{p_A}} \left(T_{E93} - T_{E52}\right) \tag{160}$$

$$V_{CW} \frac{dT_{E52}}{dt} = \boxed{F_{CW}} T_{E51} - F_{CW} T_{E52} + \frac{U_c A_c}{\rho_W C_{p_W}} \left(T_{E93} - T_{E51}\right) \tag{161}$$

$$V_R \frac{dT_{E51}}{dt} = F_{CW} T_{E52} - F_{CW} T_{E51} - \frac{U_r A_r}{\rho_W C_{p_W}} \left(T_{E52} - T_a\right) \boxed{S_{FN51}} \tag{162}$$

$$\frac{dP_e}{dt} = j E_c N_c - \frac{1}{\eta_c} \boxed{P_e} \tag{163}$$

$$E_c = E^o - \frac{R\,T_{E93}}{n\,F} \ln\left(\frac{C_{H_2O}}{C_{H_2}\sqrt{C_{O_2}}}\right) - \frac{RT_{E93}}{\alpha nF} \ln\left(\frac{j + j_l}{j_o}\right) - \frac{\zeta RT_{E93}}{nF} e^{\theta j} - Q_e \tag{164}$$

$$j_o = \kappa\, nFe^{\left(-\beta\, nFE^o/RT_{E93}\right)} \tag{165}$$

$$F_c^{out} = F_c^{in} + \frac{1}{2}\frac{N_c A_m}{nF} j \tag{166}$$

$$\Delta H_{f_{H_2O_g}} = \Delta H^o_{f_{H_2O_g}} + \left(C_p^{H_2O} - C_p^{H_2} - \frac{1}{2} C_p^{O_2}\right) \left(T_{E93} - T_o\right) \tag{167}$$

$$Q_e = \gamma_1 j + \gamma_2 j T_{E93} + \gamma_3 j^2 \tag{168}$$

$$Q_h = \left(-\Delta H_{f_{H_2O_g}}\right) \frac{j}{nF} N_c - P_e + Q_e \tag{169}$$

$$V_{V81} \frac{dT_{V81}}{dt} = r_{F81} \boxed{F_{H_2}} T_{H_2}^{in} - r_{F81} F_{H_2} T_{V81} + \frac{\rho_W C_{p_W} F_{W_{V31}}}{\rho_{H_2} C_{p_{H_2}}} T_{V31} + \frac{\boxed{Q_{V81}}}{\rho_{H_2} C_{p_{H_2}}} \tag{170}$$

$$V_{E82} \frac{dT_{E82}}{dt} = r_{F81} F_{H_2} T_{V81} + \left(1 - r_{F81}\right) F_{H_2} T_{H_2}^{in} - \frac{U_h A_h}{\rho_{H_2} C_{p_{H_2}}} \left(T_{E82} - T_a\right) + \frac{\boxed{Q_{JE82}}}{\rho_{H_2} C_{p_{H_2}}} \tag{171}$$

$$V_{V31} \frac{dT_{V31}}{dt} = F_W^{in} T_W^{in} + F_{W_{V32}} T_{V32} - F_{W_{V31}} T_{V31} \tag{172}$$

$$V_{V91} \frac{dT_{V91}}{dt} = r_{F91} F_A T_A^{in} - r_{F91} F_A T_{V91} + \frac{\rho_W C_{p_W} F_{W_{V31}}}{\rho_A C_{p_A}} T_{V31} + \frac{\boxed{Q_{V91}}}{\rho_A C_{p_A}} \tag{173}$$

$$V_{E92} \frac{dT_{E92}}{dt} = r_{F91} F_A T_{V91} + \left(1 - r_{F91}\right) F_A T_A^{in} - \frac{U_a A_a}{\rho_A C_{p_A}} \left(T_{E92} - T_a\right) + \frac{\boxed{Q_{JE92}}}{\rho_A C_{p_A}} \tag{174}$$

Nomenclature

Symbol	Description	Value	Unit
$\Delta H_{f_{H_2O_g}}, \Delta H^\circ_{f_{H_2O_g}}$	enthalpy of formation of water vapor at operating and standard conditions		$\mathrm{J \cdot mol^{-1}}$
ρ_{air}, ρ_{sol}	molar density of air and solid		$\mathrm{mol \cdot cm^{3}}$
A_{cat}, A_{mem}	surface area for cathode and membrane		$\mathrm{cm^2}$
$C^{in}_{H_2O}, C_{H_2O}$	inlet and outlet concentration of water		$\mathrm{mol \cdot cm^{-3}}$
$c_i^{H_2}, c_i^{N_2}, c_i^{H_2O}$	i^{th} heat capacity coefficients for $\mathrm{H_2,\ N_2,\ H_2O}$		
$C^{in}_{N_2}, C_{N_2}$	inlet and outlet concentration of nitrogen		$\mathrm{mol \cdot cm^{-3}}$
$C^{in}_{O_2}, C_{O_2}$	inlet and outlet concentration of oxygen		$\mathrm{mol \cdot cm^{-3}}$
$C_{p_{air}}, C_{p_{sol}}$	constant heat capacity of $\mathrm{H_2,\ O_2,\ H_2O}$, air, and solid		$\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$
$C_{p_{H_2}}, C_{p_{O_2}}, C_{p_{H_2O}}$	variable heat capacity of $\mathrm{H_2,\ O_2,\ H_2O}$,		$\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$
E_{cell}, E^o	cell and open circuit potential		V
F	Faraday's constant	96485.3399	$\mathrm{C \cdot mol^{-1}}$
$F^{in}_{cat}, F^{out}_{cat}$	inlet and outlet volumetric flow at cathode		$\mathrm{cm^3 \cdot s^{-1}}$
j, j_o	current and exchange current density		$\mathrm{A \cdot cm^{-2}}$
$k_r, \alpha, \beta, \gamma$	coefficient for reaction speed, activation, exponential and limiting current density		
n	number of electrons	2	
P_e	Electricity generated		$\mathrm{W \cdot cm^{-2} \cdot s^{-1}}$
Q_h	heat generated		$\mathrm{J \cdot cm^{-2} \cdot s^{-1}}$
R	ideal gas constant	8.3144621	$\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$
r_1, r_2, r_3	coefficients for internal resistance		
r_{H_2O}	rate of formation of water		$\mathrm{mol \cdot cm^{-2} \cdot s^{-1}}$
T_{amb}, T, T_{sol}	ambient, cathode and solid temperatures		K
U	overall heat transfer coefficient		$\mathrm{W \cdot cm^{-2} \cdot K^{-1}}$
V_{cat}	volume of cathode chamber		$\mathrm{cm^3}$

References

[1] K. C. Lauzze and D. J. Chmielewski, "Power control of a polymer electrolyte membrane fuel cell," *Ind. Eng. Chem. Res.*, vol. 45, pp. 4661–4670, May 2006.